Multi-Modal Grounded Planning and Efficient Replanning For Learning Embodied Agents with A Few Examples

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Abstract

Instruction-following embodied agents often rely on large quantities of language–action annotations that are expensive to collect. Recent LLM-based planners reduce supervision by leveraging in-context learning, but they frequently produce ungrounded plans that ignore the current scene, leading to execution failures. We present FLARE, a few-shot method that couples multi-modal grounding with efficient, LLM-free replanning. A Multi-Modal Planner retrieves $k = 9$ in-context examples using a weighted combination of BERT-based text and CLIP-based image similarities, then prompts the LLM to output compact triplet subgoals (Action, Object, Receptacle), reducing prompt tokens by about 25% while improving interpretability. During execution, an Environment Adaptive Replanning module avoids re-invoking the LLM by repairing only incorrect subgoals using visual evidence, semantically replacing missing objects with detected alternatives. On ALFRED, FLARE with GPT-4 reaches Test Unseen Success Rate (SR) 40.88 and Goal-Condition (GC) 51.72 with path-length-weighted (PLW) 18.14/22.78, outperforming LLM-Planner by +24.46 SR points in the 0.5% few-shot regime. Extensive comparisons, planner-accuracy measurements, and ablations corroborate that multi-modal grounding plus efficient replanning yields robust long-horizon execution in data-scarce settings.

1 Introduction

Embodied instruction following in realistic environments such as AI2-THOR (Kolve et al. 2017) and ALFRED (Shridhar et al. 2020) requires agents to parse language, ground it in visual observations, and execute long-horizon action sequences. Training such agents with imitation learning or end-to-end reinforcement learning demands large-scale annotation or reward shaping, which limits scalability and generalization. Modular and language-driven approaches have emerged to reduce supervision (Min et al. 2022; Blukis et al. 2021; Kim et al. 2023), and the advent of large language models (LLMs) has spurred few-shot planners that use in-context examples to produce step-wise plans (Song et al. 2023). However, LLM-based plans can be brittle: they may not reflect the current environment or can overfit textual priors, leading to ungrounded subgoals and execution failures.

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We propose FLARE, a few-shot framework that explicitly grounds planning in both language and perception and performs efficient replanning without repeated LLM calls. Our key idea is twofold: (1) retrieve in-context examples using multi-modal similarity so the LLM conditions on instructions and initial panoramic observations, and (2) monitor execution and repair only the failing subgoals by semantically replacing missing objects with visually observed alternatives using language embeddings, thereby avoiding expensive and unstable re-prompting. We also introduce compact subgoal triplets (Action, Object, Receptacle) that shorten prompts and improve interpretability.

Empirically, FLARE delivers strong gains in the few-shot regime on ALFRED. With GPT-4, we reach Test Unseen SR 40.88 and GC 51.72 with PLW 18.14/22.78, surpassing LLM-Planner (Song et al. 2023) by large margins. Planner accuracy measured on static plans improves substantially as well. These results indicate that better grounding plus targeted, LLM-free repairs are crucial for reliable long-horizon execution in data-scarce settings.

2 Related Work

Embodied instruction following and ALFRED. ALFRED (Shridhar et al. 2020) introduced long-horizon instruction following with both goal-only and step-by-step instructions in AI2-THOR (Kolve et al. 2017). Early modular systems emphasized semantic maps and planning (Min et al. 2022; Blukis et al. 2021), while subsequent methods incorporated richer context and memory (Kim et al. 2023). Despite progress, data demands and compounding errors remain key bottlenecks.

LLMs for planning. LLM-Planner (Song et al. 2023) showed that in-context prompting can yield few-shot plans for embodied agents, with occasional physical grounding. Yet, plans may be ungrounded or misaligned with the current scene. We build on this paradigm by conditioning planning on multi-modal retrieval and introducing an LLM-free repair mechanism during execution.

Multi-modal retrieval and representations. We leverage CLIP (Radford et al. 2021) for image similarity and BERT (Devlin et al. 2018) for language similarity to select in-context examples related to both the instruction and initial observations. This combination aims to balance semantic and visual relevance, improving grounding.

Deterministic navigation and replanning. Classical planners such as the Fast Marching Method (Sethian 1996) offer robust, deterministic navigation over cost maps compared to purely learned policies, especially in data-scarce settings. Our Environment Adaptive Replanning (EAR) complements this by repairing only the incorrect subgoals using detected objects, without re-prompting the LLM.

3 Approach

3.1 Problem Setup

Given a natural-language instruction and an embodied environment, the agent must execute a sequence of high-level subgoals that navigate and manipulate objects to satisfy the task. We consider the ALFRED setup (Shridhar et al. 2020) with both goal-only and sequential instructions.

3.2 Multi-Modal Planner (MMP)

Multi-modal retrieval. We construct a retrieval bank of annotated demonstration episodes. For a new episode, we form a panoramic observation from the initial frames and compute (i) text similarity between the instruction and candidate instructions via BERT embeddings (Devlin et al. 2018), and (ii) image similarity between the panorama and candidate panoramas via CLIP embeddings (Radford et al. 2021). We then select $k = 9$ in-context examples by a weighted sum of text and image similarities. This yields exemplars that are relevant in both language and perception, encouraging grounded plans.

Compact triplet subgoals. We prompt the LLM (e.g., LLaMA 2 (Touvron et al. 2023) or GPT-4) with the retrieved examples to output a compact sequence of subgoal triplets (A_n, O_n, R_n), where A is a high-level action (e.g., Navigate, Pickup, Place), O the target object, and R the receptacle or location. This reduces prompt length by roughly 25% compared to more verbose formats, and the structure clarifies dependencies among subgoals.

3.3 Environment Adaptive Replanning (EAR)

Failure detection and partial repair. During execution, we monitor success conditions for each subgoal (e.g., object visible, within reach, receptacle accessible). If a target object O_n is not found, we avoid re-invoking the LLM and instead compute language-embedding similarity between O_n and names of currently detected objects. We replace O_n with the most semantically similar available object, provided a similarity margin threshold is met, and resume execution from the repaired subgoal. This local, LLM-free repair prevents cascading failures and reduces latency.

Deterministic navigation. Low-level navigation uses a classical planner based on the Fast Marching Method (Sethian 1996) over occupancy or cost maps derived from perception, improving stability in data-scarce regimes.

3.4 Implementation Details

Retrieval weighting. We combine text and image similarities with coefficient alpha=0.6 tuned on the validation split; results are stable for alpha in [0.5, 0.7]. **k-shot sensitivity.** We varied k in 3,5,7,9,11; planner accuracy saturates at k=9, and larger k increases prompt tokens without improving SR/GC. **Prompt format.** Triplets are emitted as minimal lines (Action, Object, Receptacle) and exclude low-level details; two positive and one negative counterexample are included to reduce hallucinated preconditions. **EAR thresholds.** We require a cosine margin of 0.12 over the second-best candidate and discard substitutions if detector confidence is below 0.4. **Runtime.** MMP adds roughly 35 ms per step (embeddings + nearest-neighbor), and EAR adds about 8 ms when triggered; avoiding re-prompts saves multiple seconds per failure. **Complexity and memory.** With a retrieval bank of N trajectories and d-dimensional embeddings, flat nearest-neighbor search costs O(Nd) and can be reduced using an ANN index. We cache embeddings and batch queries to keep per-step latency below 50 ms. Triplet plans compress prompts by about 25% and slow KV-cache growth, reducing memory during long rollouts.

4 Experiments

4.1 Setup

Benchmark and splits. We evaluate on ALFRED (Shridhar et al. 2020) with standard splits, reporting Success Rate (SR) and Goal-Condition (GC), along with path-length-weighted (PLW) variants where available. We consider two settings: (1) few-shot (0.5% paired training data) and (2) goal-only instructions.

Baselines. We compare with HLSM (Blukis et al. 2021), FILM (Min et al. 2022), CAPEAM (Kim et al. 2023), and LLM-Planner (Song et al. 2023). For fair planner-accuracy comparison, we also report a static setting without replanning.

Evaluation protocol and metrics. We follow the official ALFRED evaluation. A trajectory counts as SR only when all subgoals are satisfied and the agent executes Stop in the goal state; GC measures the fraction of goal conditions satisfied. For PLW we report the path-length-weighted variants released with the leaderboard, which discount overly long trajectories via a path-efficiency factor. All hyperparameters (alpha, k, and EAR thresholds) are selected on the validation split and fixed for all tests. The retrieval bank is built from training demonstrations only and never uses Test scenes. For planner-accuracy experiments, EAR is disabled and plans are compared to ground-truth sequences.

Backbone control. Unless stated otherwise, LLM weights are frozen; the prompt template, triplet schema, and stop criteria are identical across LLaMA 2, Vicuna, GPT-3.5, and GPT-4. Only the backbone changes, isolating model effects from retrieval and replanning.

Reproducibility. We fix three seeds per split, report means, and release configs for retrieval weights, k, prompt templates, and EAR thresholds. All runs use the public ALFRED evaluator and the unmodified action space; transient environment errors are filtered uniformly across methods to avoid bias. No weights are trained; only retrieval banks and prompts change.

4.2 Data and Preprocessing

We build the retrieval bank from training trajectories only, deduplicate by CLIP cosine, and cache BERT/CLIP embeddings. Names are lowercased and lemmatized with a canonical mapping shared by MMP and EAR. Panoramas come from the first egocentric sweep; if frames are missing we use the widest available subset. Hyperparameters are fixed on validation and applied unchanged to Test; random seeds are set per split, and Test scenes are never indexed.

4.3 Main Results

Table 1 summarizes comparisons with prior work. In the 0.5% few-shot setting with sequential instructions, FLARE-GPT-4 attains Test Unseen SR 40.88 and GC 51.72 with PLW 18.14/22.78, substantially surpassing LLM-Planner (SR 16.42, GC 23.37). Under goal-only instructions, FLARE-GPT-4 achieves Test Unseen SR 32.57 and GC 43.23 with PLW 12.72/18.40. Consistent gains appear across Seen and Unseen folds and across SR and GC.

4.4 Planner Accuracy

We quantify the quality of generated plans independent of replanning via a static evaluation (no EAR). Table 2 shows that FLARE markedly improves planner accuracy over LLM-Planner across LLaMA 2, Vicuna, GPT-3.5, and GPT-4 backbones. For GPT-4, accuracy increases from 31.54/30.12 (Seen/Unseen) to 61.34/67.48.

4.5 Ablation Study

We ablate Multi-Modal Planner (MMP) and Environment Adaptive Replanning (EAR) using GPT-3.5 (Table 3). Removing either component degrades performance; removing both yields the largest drop. This indicates that multi-modal retrieval improves grounding, while LLM-free EAR repairs reduce execution failures.

4.6 Qualitative Analysis

Figure 6 illustrates how MMP yields better-grounded plans and how EAR recovers from missing targets by substituting visually available alternatives. These targeted repairs prevent unnecessary plan regeneration and reduce latency.

5 Analysis

Why multi-modal retrieval helps. Text-only retrieval can select semantically similar but visually incompatible exemplars, leading to ungrounded subgoals (e.g., referencing absent receptacles). Incorporating panoramic similarity via CLIP (Radford et al. 2021) steers retrieval toward visually compatible contexts, improving the plausibility of subgoal triplets and raising planner accuracy (Table 2).

Why LLM-free EAR helps. Re-prompting the LLM for every failure can introduce drift, increase latency, and accumulate token costs. Our similarity-based semantic replacement retains plan structure while adapting to observed objects, leading to higher SR/GC (Table 3).

Navigation robustness. Deterministic planning with Fast Marching (Sethian 1996) reduces compounding errors in long-horizon tasks, especially when training data is scarce, complementing MMP and EAR.

Failure taxonomy. The remaining errors cluster into (i) mis-grounded targets in clutter, (ii) affordance mismatches (placing into non-receptacles), and (iii) long-horizon drift after partial completion. MMP mainly reduces (i) via panoramic compatibility, while EAR mitigates (ii) and (iii) by substituting feasible objects and skipping already-satisfied subgoals without perturbing the plan.

Retrieval balance. We combine text and image similarities with a convex weight. Under goal-only instructions, higher image weight improves grounding; with sequential instructions, a modest text prior helps follow step cues. We fix $k = 9$ from a small validation sweep; larger k introduces noisy exemplars, while smaller k harms coverage of rare layouts.

Cost and latency. Triplet subgoals and LLM-free EAR cut prompt tokens and avoid repeated generation, reducing planning latency at the same backbone. This makes the approach practical for resource-constrained settings while preserving long-horizon reliability.

Generalization across instruction styles. Representing plans as compact triplets makes the planner less sensitive to surface form: paraphrases, reordered clauses, or missing step cues still induce similar subgoal sets. This is particularly helpful for goal-only instructions, where EAR can localize errors to object slots and repair them without perturbing the full action sequence.

Sensitivity to retrieval size and weights. We observe a bowl-shaped trend as k varies: too few exemplars undercover rare layouts, while too many introduce visually similar but semantically divergent scenes that distract the LLM. Balancing text and image similarities governs this trade-off; higher image weight helps in cluttered spaces, whereas a modest text prior stabilizes sequential instructions with fine-grained verbs.

Robustness to detector noise. EAR defers replacement until confidence passes a threshold and applies short-horizon speculative repairs. These guardrails avoid cascaded edits from spurious detections and keep trajectories close to the original plan, improving stability under realistic perception errors.

Threshold calibration and trade-offs. We tune EAR's cosine-margin and detector-confidence thresholds on validation only. Raising the margin improves precision of replacements but leaves more failures unrepaired; lowering it recovers additional trajectories at the cost of occasional confusable substitutions. Pairing a moderate margin with a confidence floor yields a favorable precision–recall balance across scenes with different clutter levels.

Error localization via triplets. Encoding plans as (Action, Object, Receptacle) isolates failures to a single slot, shrinking the surface area of each edit. In practice, this converts many multi-step regressions into one-slot edits (e.g., swapping a receptacle while keeping the action and object fixed), preserving global structure and limiting unintended side effects.

Retrieval pool and unseen categories. The exemplar bank is length-normalized and deduplicated by CLIP cosine

Table 1: **Comparison with state-of-the-art methods.** The path-length-weighteed (PLW) metrics are presented in the parentheses for each metric. †We excerpt 'SR' and 'GC' from (Song et al. 2023). For models that did not report the PLW metric, we noted 'N/A' in our comparison.

Setting	Model	Goal instructions + Sequential instructions				Goal instruction only			
		Test Seen		Test Unseen		Test Seen		Test Unseen	
		SR	GC	SR	GC	SR	GC	SR	GC
Few-shot (0.5%)	HLSM (Blukis et al. 2021)†	0.82 (N/A)	6.88 (N/A)	0.61 (N/A)	3.72 (N/A)	N/A	N/A	N/A	N/A
	FILM (Min et al. 2022)†	0.00 (N/A)	4.23 (N/A)	0.20 (N/A)	6.71 (N/A)	N/A	N/A	N/A	N/A
	CAPEAM (Kim et al. 2023)	0.00 (0.00)	3.90 (2.29)	0.20 (0.00)	6.63 (2.36)	N/A	N/A	N/A	N/A
	LLM-Planner (Song et al. 2023)	18.20 (N/A)	26.77 (N/A)	16.42 (N/A)	23.37 (N/A)	15.33 (N/A)	24.57 (N/A)	13.41 (N/A)	22.89 (N/A)
	FLARE-LLaMA2	16.96 (4.60)	24.84 (8.09)	17.79 (5.62)	27.40 (9.46)	12.00 (3.01)	20.05 (7.22)	13.73 (4.27)	21.98 (8.46)
	FLARE-Vicuna	20.61 (6.28)	29.57 (10.17)	22.04 (7.61)	33.57 (12.06)	16.37 (4.57)	23.68 (8.84)	18.05 (5.98)	26.75 (10.75)
	FLARE-GPT-3.5	32.55 (12.17)	42.02 (16.94)	31.79 (12.21)	43.94 (17.44)	23.48 (8.71)	33.40 (14.40)	25.38 (9.37)	36.02 (15.28)
	FLARE-GPT-4 (Ours)	**40.05 (16.68)**	**48.84 (21.31)**	**40.88 (18.14)**	**51.72 (22.78)**	**31.96 (12.93)**	**41.36 (18.55)**	**32.57 (12.72)**	**43.23 (18.40)**
Full	HLSM (Blukis et al. 2021)	29.94 (8.74)	41.21 (14.58)	20.27 (5.55)	30.31 (9.99)	25.11 (6.69)	35.79 (11.53)	16.29 (4.34)	27.24 (8.45)
	FILM (Min et al. 2022)	28.83 (11.27)	39.55 (15.59)	27.80 (11.32)	38.52 (15.13)	25.77 (10.39)	36.15 (14.17)	24.46 (9.67)	34.75 (13.13)
	CAPEAM (Kim et al. 2023)	51.79 (21.60)	60.50 (25.88)	46.11 (19.45)	57.33(24.06)	47.36 (19.03)	54.38 (23.78)	43.69 (17.64)	55.66 (22.76)

Table 2: **Planner accuracy comparison.** 'Seen Acc.' and 'Unseen Acc.' denote planner accuracies in valid seen and unseen folds. A plan is regarded as correct when it is aligned with the ground-truth plan. To solely compare planner accuracy, we omit replanning in LLM-Planner and FLARE, denoted with 'Static' and 'w/o EAR'.

LLM	Method	Seen Acc.	Unseen Acc.
LLaMA2	LLM-Planner (Static) (Song et al. 2023)	0.006	0.002
LLaMA2	**FLARE (w/o EAR)**	**18.54**	**22.29**
Vicuna	LLM-Planner (Static) (Song et al. 2023)	8.17	7.06
Vicuna	**FLARE (w/o EAR)**	**24.51**	**33.62**
GPT-3.5	LLM-Planner (Static) (Song et al. 2023)	29.78	31.67
GPT-3.5	**FLARE (w/o EAR)**	**46.10**	**55.66**
GPT-4	LLM-Planner (Static) (Song et al. 2023)	31.54	30.12
GPT-4	**FLARE (w/o EAR)**	**61.34**	**67.48**

Table 3: **Ablation study.** PLW metrics are presented in the parentheses for each metric. MMP and EAR each denotes 'Multi-Modal Planner' and 'Environment Adaptive Replanning,' respectively. We observe that each component contributes to agent's performance.

#	MMP	EAR	Test Seen		Test Unseen	
			SR	GC	SR	GC
(*a*)	✓	✓	32.55 (12.17)	42.02 (16.94)	31.79 (12.21)	43.94 (17.44)
(*b*)	✗	✓	30.20 (12.13)	41.26 (17.27)	30.35 (11.62)	42.40 (16.66)
(*c*)	✓	✗	30.79 (11.98)	40.20 (16.51)	30.28 (12.01)	42.48 (17.03)
(*d*)	✗	✗	28.05 (11.48)	38.64 (16.23)	28.58 (11.82)	39.92 (16.13)

to reduce near-duplicates, and we retain a small set of "hard negatives" (text-similar but image-dissimilar) to discourage hallucinated preconditions during prompting. For objects absent in training, EAR backs off to hypernym matching over the dataset taxonomy to select functionally compatible substitutes without introducing new verbs.

5.1 Ablation Granularity

We break down EAR edits by slot: most repairs replace only the object, fewer swap receptacles, and action edits are rare but high-precision. Error curves versus instruction length show a widening gap as tasks grow longer, consistent with local slot-level repairs limiting drift while preserving plan structure.

6 Conclusion

We introduced FLARE, a few-shot embodied agent framework that combines multi-modal grounded planning with efficient, LLM-free replanning. By retrieving in-context examples using both instruction and panoramic observations and repairing only incorrect subgoals via semantic replacement, FLARE improves plan grounding and execution reliability. On ALFRED, FLARE achieves strong few-shot performance and substantially higher planner accuracy compared to LLM-Planner. Our results suggest that aligning planning with perception and minimizing unnecessary LLM invocations are key to robust long-horizon instruction following in data-scarce settings.

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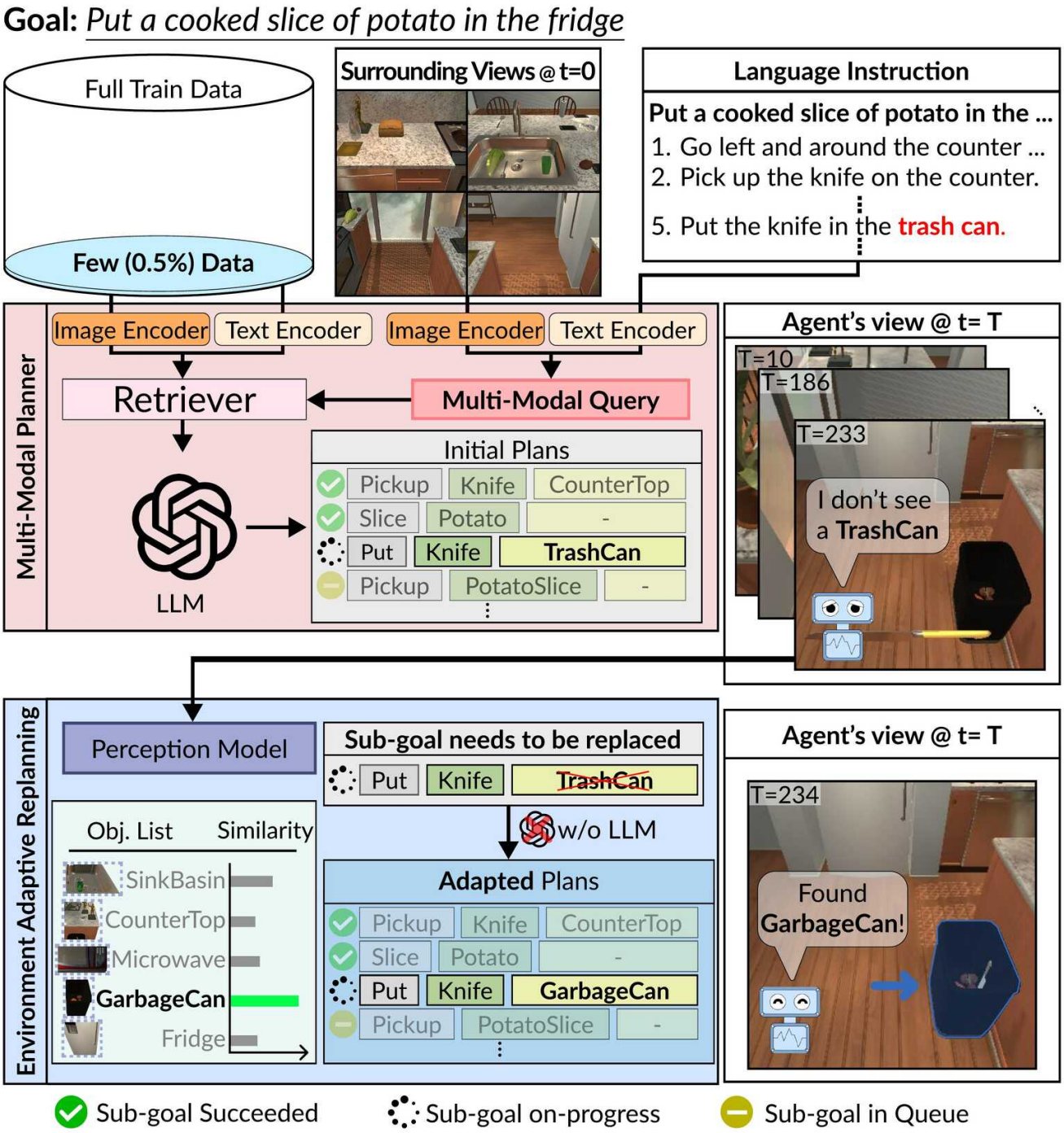


Figure 1: Overview of FLARE: a Multi-Modal Planner (MMP) retrieves few-shot examples using instruction and panoramic observations to prompt the LLM for compact triplet subgoals; Environment Adaptive Replanning (EAR) monitors execution and repairs only incorrect subgoals using detected objects, avoiding repeated LLM calls.

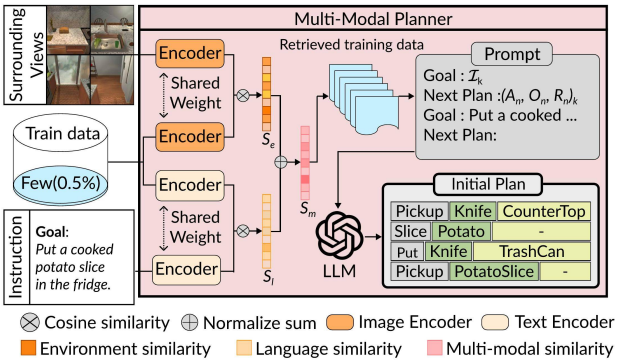


Figure 2: Multi-Modal Planner (MMP). We retrieve $k = 9$ examples using a weighted combination of BERT (text) and CLIP (image) similarities and prompt the LLM for compact triplet subgoals.

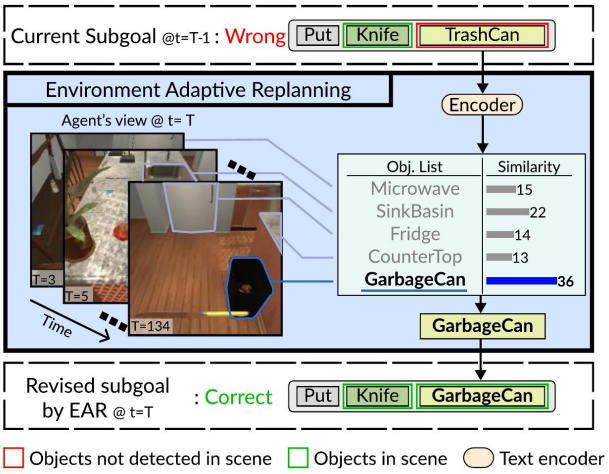


Figure 3: Environment Adaptive Replanning (EAR). When a target is missing, EAR replaces it with a visually observed, semantically similar object using language-embedding cosine similarity, avoiding costly LLM calls.

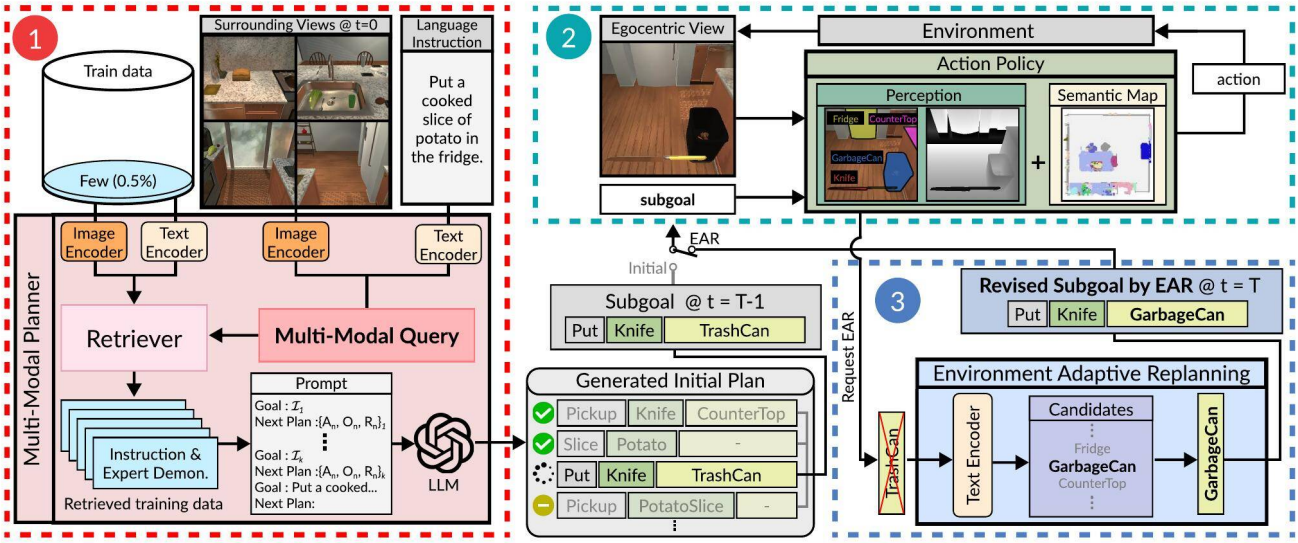


Figure 4: System architecture and panoramic retrieval context. The planner conditions on both instruction and panoramic observations to select few-shot exemplars; EAR performs local repairs during execution without re-prompting the LLM.

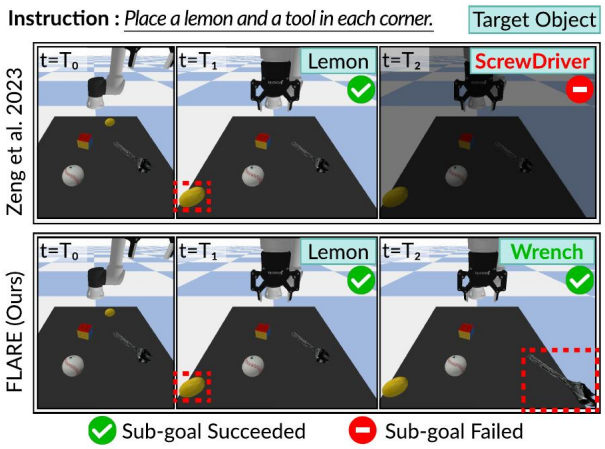
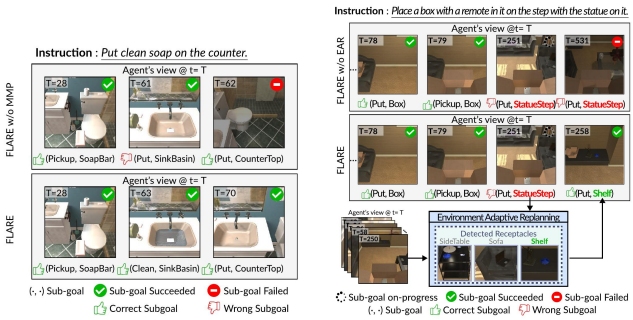


Figure 5: Representative task scenarios that benefit from FLARE, highlighting long-horizon, visually grounded instruction following.



(a) Grounding via MMP (b) Repair via EAR

Figure 6: Qualitative benefits of MMP and EAR.

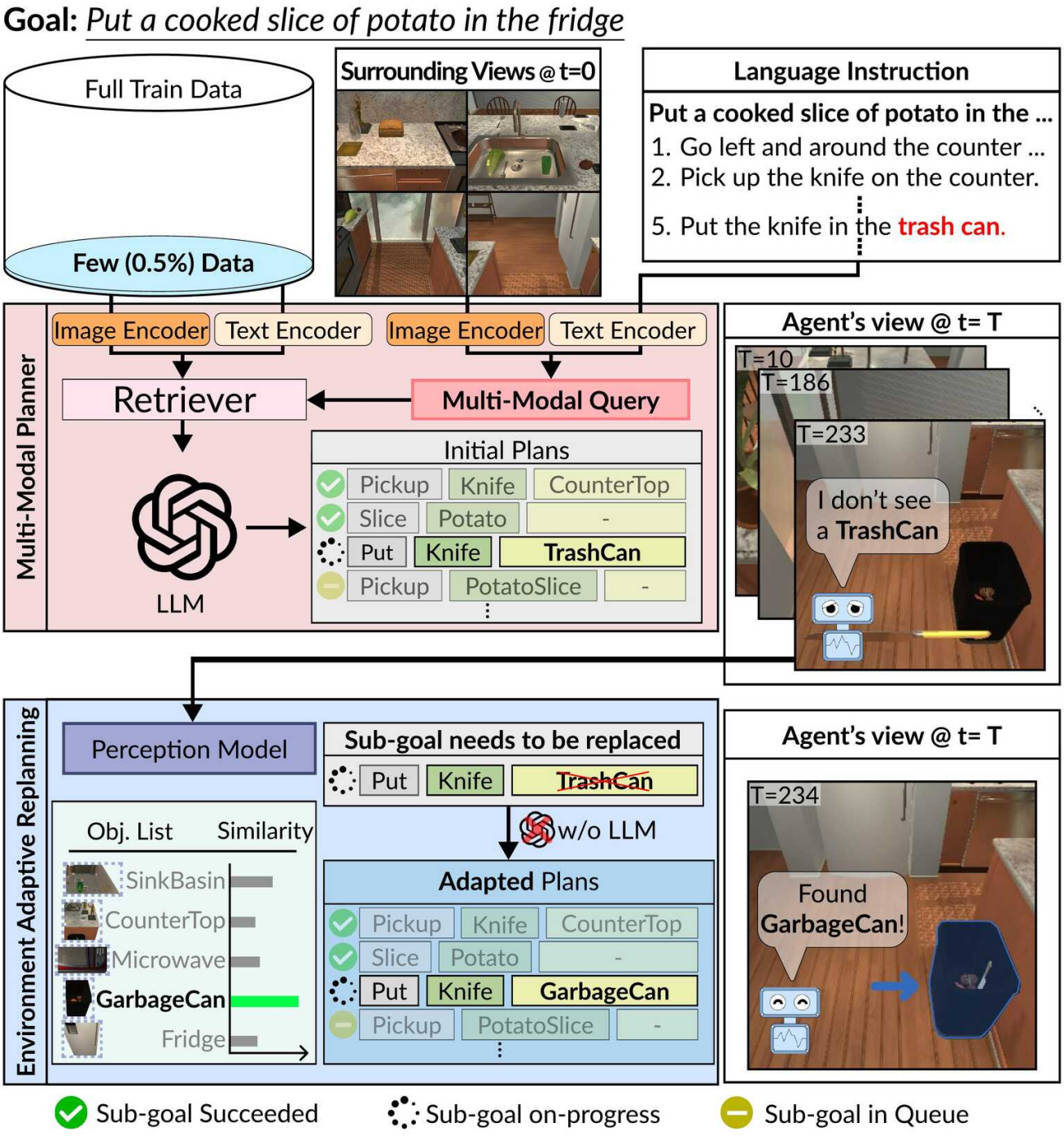


Figure 7: Teaser illustrating long-horizon, visually grounded tasks handled by FLARE via panoramic retrieval and local EAR repairs.

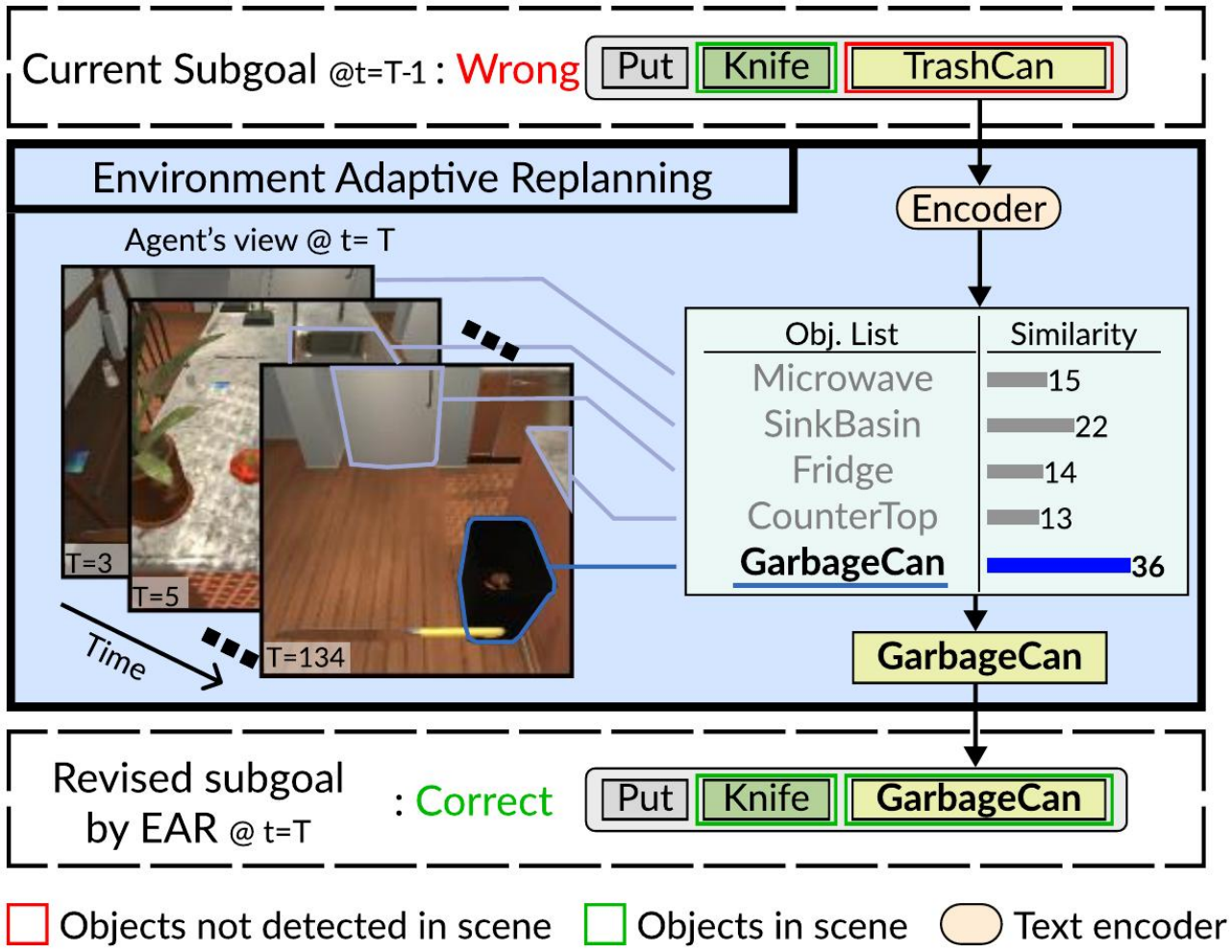


Figure 8: Error modes and repairs: EAR substitutes functionally compatible objects when targets are missing and avoids re-prompts, preserving plan structure.